

Problem 1: Find the Fourier Series solutions of the Boundary Value Problems:

(i) $y'' - 4y = -1 \quad 0 < t < 1$
 $y(0) = y(1) = 0$

(ii) $y'' + 2y = t \quad 0 < t < \pi$
 $y'(0) = y'(\pi) = 0$